Week 2 web page: delivery guide

Instructor notes for Week2_DeterministicFunctions.html

General

- Page: https://kevintshoemaker.github.io/NRES-746/Week2_DeterministicFunctions.html. Open it before class; the three widgets need no internet once loaded.
- For live demos, run the matching section of `CODE_Week2_DeterministicFunctions.R` (same seeds, same numbers). Everything runs in a second. Three lines are meant to fail (the WOD bugs and the `if` hockey stick); the script wraps them in `try()`.
- Zoom the browser to 125-150% for the projector.
- Questions on the page have no answers. Answers are below; *Land:* is where discussion should end up.
- Readings for Canvas: Bolker 3.1-3.5 and the R supplement 3.7 (printed pp. 99-138).
- WODs: *R Syntax and Function Adjustments* (Monday) and *Limits and a Derivative* (Wednesday), keys in `wod/answer_keys/`. Both are worked on the page (`#wod-syntax`, `#wod-limits`).
- Hard dependency: Lab 2 Act 2 tells students to use "the 'throw out the lower-order terms' trick" for Holling type IV. If the limits section doesn't land Monday, that question stalls.

As taught (Aug. 31-Sept. 2, 2026)

- Monday (per the announcement, added Nov. 4): Bolker ch. 3 framed as giving theories a mathematical form, with parameter estimation and uncertainty deferred (now in the `#functions` big picture). The announcement set the WOD credit rule (attempt every question, or explain what's confusing), now in the Week 1 page's `#course`.
- Wednesday (per the announcement, added Nov. 4): finished deterministic functions, with time on the logistic. Only one student reached $b/4$ in the WOD's optional part; the announcement assigned the two-forms equivalence and the max slope as practice (now a page question in `#wod-limits`) and the ch. 3 review checklist (now "Before moving on" in `#mechanism`). Stats Chats details went to the Week 1 page's `#course`.
- No other detailed report. The page was built Oct. 30, after the week, from the two PDF NOTES (now in `lecture_notes/archive/`), the two WODs, Lab 2, and Bolker ch. 3. The Wednesday WOD's hint ("we've already done the first two together") says Monday covered the Michaelis-Menten and Ricker limits.
- There were no lab-opening NOTES for Lab 2. The page's lab H1 (`#lab2`, loops, `outer()`, `replicate()`) is new; use it as a short opening or as a reference students can read on

their own.
- New on the page: the three widgets, the hockey-stick `if` example, the risk surface, the quadrat resampling example, the Euler loop for the logistic, the monomolecular comparison, the discussion prompts, and both bonuses.
- No legacy `LECTUREn` page covered this week, so nothing was archived.

Part 1: Reading a deterministic function (Monday, ~32 min)

Big picture (~3 min)

- Point at MODELS. A model = deterministic function + noise; this week is the function. Show the tadpole plot: what shape, and what would a function need to do to match it?
- Numerically (plot it) vs. analytically (limits, derivatives).

WOD (~5 min)

- Bug 1 gives 1.84 instead of 0.135: no error, just the wrong number. Bug 2: no `e` in R. Bug 3: `ricker()` runs immediately; the fix gives 0.92, 1.84, 2.76 (each aval fills `a`).
- Seasonal: $5 + 5\sin(2\pi(t - 91)/365)$. Don't name day 91's kind of point yet; it is the inflection point on Wednesday.

Writing and plotting functions (~6 min)

- `x` first, then parameters with defaults (Bolker's convention, and the one Lab 2 asks for). `curve()` with `add = TRUE`.
- "`curve(hockey_if(x), from = 0, to = 10)` fails, but `curve(hockey(x), ...)` works. What does `curve()` pass to the function, and why does that break the `if` version?" *Land:* `curve()` passes one vector of 101 x values in a single call. `if` needs one `TRUE`/`FALSE`, so a vector condition is an error; `ifelse()` works element by element.

Limits (~10 min)

- **Board:** the hierarchy, then Michaelis-Menten at both ends. Stress "which term matters," not "plug in infinity," and that b is negligible at one end and dominant at the other. Half-max at b; 90% at $9b$.
- Widget stops: Michaelis-Menten at $b = 1$, slide x from 0.001 (small-x ratio 1.001) to 1000 (large-x ratio 1.001); near $x = 1$ both are off by a factor of 2. Then Ricker: the curve plunges at large x (no power approximation).
- "Choose Holling type III... why slope 2 rather than slope 1? Where is neither approximation good, and how does that zone move when you change b?" *Land:* near zero $f \approx (a/b^2)x^2$, a power of 2. The small-x ratio is $1 + (x/b)^2$ and the large-x ratio $1 + (b/x)^2$, so both are within 10% only below $0.32b$ or above $3.2b$. The poor zone runs from about $0.3b$ to $3b$ and slides with b (same width on the log axis). The same-degree trick is what Lab 2 needs, so don't do Holling IV here.

Derivatives and critical points (~8 min)

- Ricker: $f' = a(1 - bx)e^{-bx}$, peak $1/b$, height $a/(be)$. With Lab 1's values: 12 mm, 3.97 (matches Lab 1 Act 3's table). `D()` and `optimize()` agree.
- Michaelis-Menten: $f' = ab/(b + x)^2 > 0$, monotonic. Root vs. critical point.
- "Doubling a changes the height but not the location. What does doubling b do to each? Which parameter would you change for tadpoles vulnerable over a wider range of sizes?" *Land:* doubling b halves both the location ($1/b$) and the height ($a/(be)$). A wider range means a smaller b (b is a scale parameter: it stretches the curve along x); raise a too if the peak height should stay put.
- Don't derive the flexible Ricker's critical point (Lab 2 Act 2 1b).

Discussion (~3-5 min)

- Q1 (what caps a saturating curve) *Land:* open. Prey: handling time, gut capacity; photosynthesis: enzyme capacity; recruitment: space or food. b is set by how fast the limit kicks in (attack rate, affinity).
- Q2 (quadratic survival, 0.20 at 40%, -0.40 at 60%) *Land:* polynomials go to $\pm\infty$ outside the data and know nothing about 0 and 1. A better function stays between 0 and 1 and goes to 0 at both extremes (e.g. a logistic of a quadratic, or a hump-shaped function scaled below 1).
- Q3 (numerical vs. analytical) *Land:* plots are enough to check a shape over the data's range; algebra tells you what happens outside it, what each parameter means, and where to start a search.

Part 2: Repeating a calculation in R (lab opening, optional, ~10 min)

- Use this if there is time before the Acts, or point students to it. It gives tools, not Act answers. Don't run an SSR grid search over (a, b) (Act 1) or write out a bootstrap loop (Act 3).
- Three ways to evaluate a function over a vector; `sapply()` over parameter values returns one column per value.
- "`curves` has 21 rows and 5 columns. Which goes with x and which with b?" *Land:* rows are x (21 values of `xvec`), columns are the 5 values of b. Check with `nrow(curves) == length(xvec)`, or plot one column.
- `outer()` + `image()`/`contour()` is the tool Act 1's heatmap needs.
- "Along a line of fixed attack rate, risk falls as density rises. Why? Where does density make the least difference?" *Land:* each prey caught costs handling time h, so at high density the predator is busy and each tadpole's share falls (roughly $1/(hN)$). At a low attack rate, αhN stays small and risk $\approx \alpha$: 0.0099 at $N = 1$ vs. 0.0075 at $N = 40$, compared with 0.35 vs. 0.028 at $\alpha = 0.5$.
- Quadrats: full mean 2.75; 5 random quadrats give 1.00 to 4.80 (middle 95%, seed 2).
- "How would the histogram change with 10 quadrats? What if you draw all 20 without

replacement?" *Land:* narrower with 10. With all 20 and no replacement every draw is the same data, so every mean is 2.75 and the spread is zero. Drawing with replacement makes each resample different, which is why the bootstrap does it. This touches Act 3's planning question; use it after Act 3 or leave it as reading.

Part 3: The bestiary of functions (Wednesday, ~32 min)

Big picture (~2 min)

- Still MODELS. Bolker says treat 3.5 as a map, not a list. Students have already met Michaelis-Menten, Ricker, Holling IV (lab), Holling III and the logistic (WOD).

WOD (~6 min)

- Table of limits. **Board:** Holling III at large x: drop b^2 inside the denominator, then $ax^2/x^2 = a$. This is the move Lab 2's Holling IV needed.
- Logistic: $f'(0) = b/4$, not zero. Numerical check gives 0.5 for $b = 2$.
- "Convince yourself that $e^{a+bx}/(1+e^{a+bx})$ and $1/(1+e^{-(a+bx)})$ are the same function. Then find the derivative… is the maximum slope still $b/4$?" *Land:* divide top and bottom by e^{a+bx}. $f'(x) = b\,e^{-(a+bx)}/(1+e^{-(a+bx)})^2 = bf(1-f)$; steepest where $f = 1/2$, at $x = -a/b$, with slope $b/4$ regardless of a (a shifts, b sets steepness). Good homework if short on time.

Critical points and inflection points (~10 min)

- Critical point = flattest nearby ($f' = 0$); inflection point = steepest nearby ($f'' = 0$ with a sign change). Logistic midpoint; WOD day 91; a sine has one every half cycle. Ricker: peak at $1/b$, inflection at $2/b$ ($f''(2/b) = 0$).
- Widget stops: Ricker (both buttons: peak at 2, inflection at 4, slope most negative there); logistic (inflection at 5, no critical point); seasonal sine (inflection day 91, peak day 182).
- "Choose Michaelis-Menten. Why do both buttons stay greyed out? Now choose Holling type III. What does it have that Michaelis-Menten lacks, and what does that say about how a predator treats rare prey?" *Land:* Michaelis-Menten's slope is always positive (no critical point) and it is concave down everywhere (no inflection). Holling III starts flat and concave up, with an inflection at $b/\sqrt{3} = 1.15$: at low density the predator takes few prey (switching to commoner prey, refuges), then feeding accelerates.
- Gompertz: inflection at $1/e \approx 0.37$ of the way up.

Map of the bestiary (~8 min)

- Walk the table by family, one sentence each on pros and cons. Mention Bolker's Holling IV (x^2 in the numerator) vs. the Andrews/Monod-Haldane version (x).
- Widget stops: Michaelis-Menten (default), then Ricker, logistic, Gompertz, power law with $b < 0$, Shepherd.

- "Choose Shepherd and move c from 0.5 to 3. For which values does it peak? What does it become at $c = 1$? What would a fitted c well above 1 say for tadpoles?" *Land:* a peak only when $c > 1$, at $(b/(c-1))^{1/c}$ (1.41 at the defaults); $c = 1$ is Michaelis-Menten; $c < 1$ never levels off. $c \gg 1$ means kills fall steeply for large tadpoles (they escape or are too big to handle).

Where shapes come from (~6 min)

- Logistic from $dn/dt = rn(1 - n/K)$; the Euler loop matches the formula; fastest growth at $K/2 = 50$.
- Michaelis-Menten as Holling II: asymptote $1/h$, initial slope α.
- Monomolecular (asymptote 5) vs. closest Michaelis-Menten ($a = 5.93$, $b = 2.11$): within 0.26 everywhere on $[0, 15]$.
- "A manager asks for the maximum feeding rate. What do you tell them, and what data would settle it?" *Land:* the data can't say: 5 or 5.93 depending on the function, a 19% difference from a choice the data couldn't make. Report the uncertainty from the model choice, and get data at higher densities, where the curves separate, or use a mechanism (handling time) to choose.

Discussion (~3-5 min)

- Q1 (adding a parameter) *Land:* gain: peak location and width can vary separately, so the curve can match a sharp hump. Lose: with few data points the extra parameter is poorly estimated, and a better fit to these data may predict new data worse (Lab 2 Act 2's 2d and Act 3 make this concrete).
- Q2 (inflection point as "optimal temperature") *Land:* no. The inflection is where germination is most sensitive to temperature. A logistic is monotonic, so it has no optimum; an optimum needs a hump, a critical point where germination falls again at high temperature.
- Q3 (thresholds) *Land:* open. A hockey stick is an abrupt switch, a steep logistic a fast but smooth one; telling them apart needs data close to the threshold.

Bonuses (skip if behind)

- Taylor: Ricker $\approx ax - abx^2$ near 0; logistic $\approx 1/2 + (b/4)x$ near its midpoint ($f''(0) = 0$). Preview of quadratic approximations for likelihood intervals.
- Shift and scale: the shifted Michaelis-Menten starts at $c = 3$ from $d = 1$, reaches 3 at $x = 5$, approaches 5. Reparameterizing and linearizing transformations (semi-log, log-log, logit).